

**INVESTIGATING THE DYNAMICS OF A CHAOTIC
SPROTT CIRCUIT**

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MARCH 2004

Investigating the Dynamics of a Chaotic Sprott Circuit

by

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An Undergraduate Thesis Submitted to the
National Institute of Physics
College of Science
University of the Philippines
Diliman, Quezon City 1101

As Partial Fulfillment of the Requirements
for the Degree of
BACHELOR OF SCIENCE IN APPLIED PHYSICS

24 March 2004



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"Investigating the Dynamics of A Chaotic Sprott Circuit"

in partial fulfillment of the requirement for the degree of Bachelor of Science in Applied Physics

on

Wednesday, 24 March 2004
NIP Audio-Visual Room
3:00 P.M.

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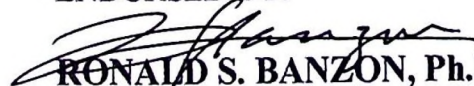
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Dear Dr. Saloma:

Below is the official action of the panel members in the undergraduate thesis defense of Mr. Alfie R. Pelicano. The defense of thesis entitled "**Investigating the Dynamics of a Chaotic Sprott Circuit**" was held at PH 3213, Wednesday, 24 March 2004, 3:00 P.M.

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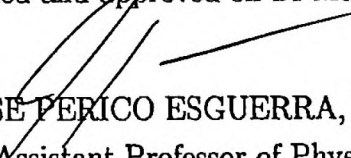
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Sincerely yours,


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This is to certify that this undergraduate thesis entitled "Investigating the Dynamics of a Chaotic Sprott Circuit", submitted by ALFIE R. PELICANO to fulfill part of the requirements for the degree of Bachelor of Science in Applied Physics, was successfully presented, defended and approved on 24 March 2004.



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Acknowledgements

For the thesis problem, the guidance, the encouragement, the confidence, the unwritten rules, the food, the faith, the jokes, and, um, the money I borrowed and still owe – thanks for it all, Sir Perry. I'll pay any remaining balance I may have forgotten as soon as I get my first paycheck.

To Erwin and Johnrob for being the panelists in my thesis defense. Thank you for not making my defense as bloody as I had imagined it to be.

To Ma'am Jing for the last minute consultations regarding the FFT I used for my thesis, the FFT codes themselves, the occasional rides home late at night and for treating me and Tra to Trellis last April 3, 2004. I'll be watching out for Cecille DeMille's "Ten Commandments in Drag" this Holy Week. A romantic angle? Really?

To my parents, for their patience and endless support, and for being there even if they rarely understood me. You know I love you two... No amount of words can ever say that. I really should pay for all the phone bills I piled up over the past year... Come to think of it, especially for the past year.

To my brother and sister, for everything that they are to me. To my sister specifically for the computer – without which I never would've finished the first draft – and for dropping me off at NIP on the day of my defense. To my brother for, well, offering to drop me off at NIP the night before. It's the thought that counts, I suppose.

To Theory – for the coffee fetish, the insanity, the music, the lessons, the diversity, the dishwashing therapy, the "fountain of youth", the fun, the food, the SysAd experience (or is it the SysAd challenge?), the life, the aircon and the embodiment of chaos...

To Tel, for helping me choose my very glossy defense outfit. Really, Tel, I like blue,

but aquamarine really isn't me. And it practically glows in the dark, too.

To Tra for telling me not to screw up the defense, because otherwise it would be a total letdown considering you and Tel chose such a great outfit for me (...)

To Sel, for wearing a colorful cap on the day of my defense. You were a beacon of distraction, er, light during the defense. You'll always be my baby sister.

To Tel, Tra and Sel, collectively. I love you guys. There's a lot of trauma I'm blocking out right now, though, so I'll just leave it to Tel and Tra to jointly create vivid false memories of all the adventures – or otherwise – we've shared together.

To Bhazel who, um, *behaved* in the background. Well, you know what I mean. For trusting me, for not holding back, for making me a part of your life... thank you. I still am a part of your life, by the way, and that part will never change. I love you.

Investigating the Dynamics of a Chaotic Sprott Circuit

Abstract

In this thesis, we numerically investigate the effect of introducing a sinusoidally varying resistance to a modified Sprott circuit which is known to exhibit chaos. The resulting dynamics was characterized using a standard technique where the systems Lyapunov exponents are computed using an algorithm created by Alan Wolf.

In varying the magnitude and frequency of the added time-dependent resistance, it was found that for small frequency values, the systems maximal Lyapunov exponent exhibited an overall gradual fluctuation which decreased in amplitude as the magnitude of the resistance was decreased. For higher frequencies of the resistance, the systems maximal Lyapunov exponent settled to approximately constant positive values which also decrease as the magnitude of the resistance was decreased.

Contents

Abstract	iv
1 Introduction	1
2 Standard Nonlinear Dynamics Tools	5
2.1 Phase Space Trajectories	5
2.2 Power Spectrum	7
2.3 Lyapunov Exponents	8
3 Sprott Circuits	10
4 Methodology	12
5 Results and Discussion	15
6 Conclusions and Recommendations	23
A BASGEN (dataBASE GENERator)	25
B FET (Fixed Evolution Time)	30
C FFT (Fast Fourier Transform)	37

CONTENTS

vi

Bibliography

39

List of Figures

1.1	General form of Chua's circuit, where the component labeled RNL is the piecewise linear resistor (Taken from reference [10]).	2
1.2	Sprott's general circuit for solving equation 1.1 which uses one of the feedback elements in figure 1.3 for $G(x)$. Above or to the left of each component are values that would make the circuit behave in real time, while values not shown are unity. Below or to the right are practical values for $A = 0.6$ that produce similar outputs for the op amps and scale the frequency up by a factor of 10^4 into the audio range ($f = 10^4/2\pi \approx 1592Hz$) (Taken from reference [3]).	3
1.3	Some nonlinear forms for $G(x)$ that produce chaos and circuits to produce them. The parameter B is selected to be well within the chaotic regime. Constant C determines the size of the attractor and has been adjusted in each case so that x reaches a maximum value of slightly less than $5V$, a voltage well above noise level, but comfortably below a value that would typically saturate the op amps (Taken from reference [3]).	4
2.1	Phase space plots of the harmonic oscillator for different $\frac{k}{m}$ values.	6

2.2	Graphic representation of the delay reconstructed orbit from a single time series (Taken from reference [9]).	8
3.1	Phase space plot for one of the parameter pairs examined in the work that yield the 'most chaotic' solutions.	10
4.1	Chaotic circuit implementation of eqn. 4.1 with $R = 0\Omega$, using inverting op amps. The diodes are germanium, the battery is 1V, the capacitors are $0.1\mu F$, and the resistors are $1k\Omega$ except for the variable resistor – related to the parameter A in eqn. 4.1 via $R_{variable} = \frac{1}{A}$ – which should be adjustable from 1 to $2k\Omega$ (Taken from reference [4]).	13
5.1	System dependence on ω for $(0.15\Omega, \omega)$ pair.	15
5.2	System dependence on ω for $(0.05\Omega, \omega)$ pair [dots], superimposed on $(0.15\Omega, \omega)$ pair [solid lines] for comparison.	16
5.3	System dependence on R for large ω	17
5.4	Enlarged view of the system's varying chaoticity as a function of ω for small ω values.	18
5.5	"Averaging behaviour" of the system for large ω values.	19

Chapter 1

Introduction

The phenomenon of chaos has been a much studied area in the field of nonlinear dynamics due in large part to Lorenz's formulation in 1963 of three coupled first order ordinary differential equations (ODE's) modelling the dynamics of atmospheric convection. The fervor that followed resulted in a widespread study that explored increasingly more complex dynamical models to explain and predict physical phenomena in fields ranging from mathematics, physics, biology, and chemistry, to engineering, economics, and medicine. Although seemingly ubiquitous in such a variety of fields, chaos itself has only been recently explored to such a degree due to the previous successes of linear models in the sciences and the delayed advent of powerful computers. Also, that it went largely unnoticed for so long can be attributed to the relative rarity of chaos in systems described by low-dimensional ODE systems [1].

Before going further, a brief overview of chaotic behaviour is in order. Though lacking in a commonly accepted definition, widely accepted characteristics of chaos include [2]:

1. a non-periodic bounded long-time evolution of a time-continuous dynamical system leading to a strange attractor in its state space;
2. a sensitive dependence on the initial conditions being characterized by the appearance of at least one positive Lyapunov exponent;
3. a phase space dimension $n \geq 3$ for autonomous (i.e., no explicit time dependence) time-continuous dynamical systems $\dot{\vec{x}} = \vec{F}(\vec{x})$ with $\vec{x}(t) = (x_1(t), \dots, x_n(t))$ and;
4. some form of nonlinearity in the vector field $\vec{F}(\vec{x})$.

The first criteria precludes the existence of unbounded solutions which may be mistaken for chaotic behaviour when the second is taken into account. This means

that given the fulfillment of the first criteria, all positive Lyapunov exponents can only mean the existence of chaos in the system. The third requirement is a consequence of the Poincaré-Bendixson theorem [5], which implies the necessity of three variables in a dynamical system, stating explicitly that the long-time limit of any “smooth” two-dimensional flow is either a fixed point or a periodic solution.

The simplest systems which can be easily implemented for displaying and studying chaos on a physical level are chaotic circuits, simplest, that is, in the implementation of the experimental setup. In fact, an electronic circuit designed by Leon Chua in 1983 consisted one of the first demonstrations which showed that chaos was not merely a mathematical abstraction. Known as Chua’s circuit, it consists of a piecewise linear resistor as its nonlinearity, with two capacitors, one resistor and one inductor (figure 1.1). The circuit is unforced (autonomous). Its chaotic aspects, i.e., bifurcation values, Lyapunov exponents, various dimensions, etc., have been extensively studied in the literature both experimentally and theoretically [8].

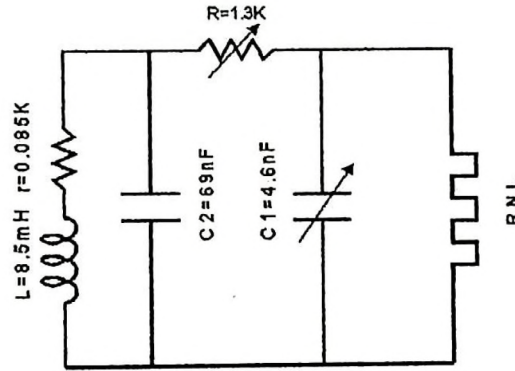
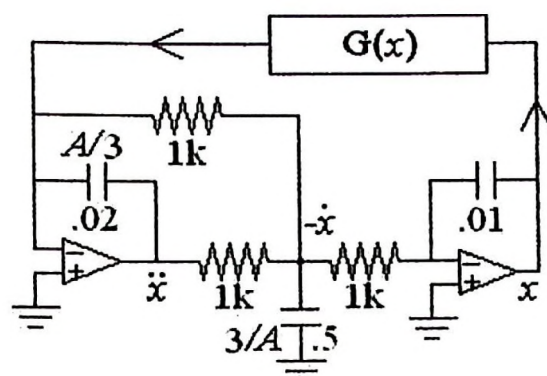


Figure 1.1: General form of Chua’s circuit, where the component labeled RNL is the piecewise linear resistor (Taken from reference [10]).

In 2000, Julien Clinton Sprott investigated a series of chaotic circuits of the form

$$\frac{d^3x}{dt^3} + A\frac{d^2x}{dt^2} + \frac{dx}{dt} = G(x) \quad (1.1)$$

where $G(x)$ is one of a number of elementary piecewise linear functions some of which are given by figure 1.3. The circuit implementation itself can be seen in figure 1.2. The value of $x(t)$ and its derivatives are voltages sampled by tapping specific nodes on the circuit where voltage is integrated using op amp circuits with integrator configurations or analog integrator circuits. It seems little work has been done on the system given by equation 1.1 with a nonlinear $G(x)$. Coulett, Tresser, and Arneodo observed chaos in



the same system via numerical simulations with a cubic and a special piecewise linear form of $G(x)$, while Rul'kov et. al. devised an RLC circuit with an unspecified nonlinear amplifier to produce a particular form of $G(x)$. Sprott goes on to list several functions $G(x)$ with the typical parameter values for which chaos occurs. [3, 4]

In this thesis, we will numerically examine a particular chaotic circuit designed by Sprott to realize an absolute value nonlinearity $G(x) = |x| - 1$ after introducing a modification in the form of a sinusoidally varying resistance. Solutions of the circuit's equivalent ODE is obtained via a standard 4th order Runge-Kutta algorithm while its dynamics is characterized by using an algorithm developed by Wolf [6, 7] for obtaining the Lyapunov exponent of a given time series.

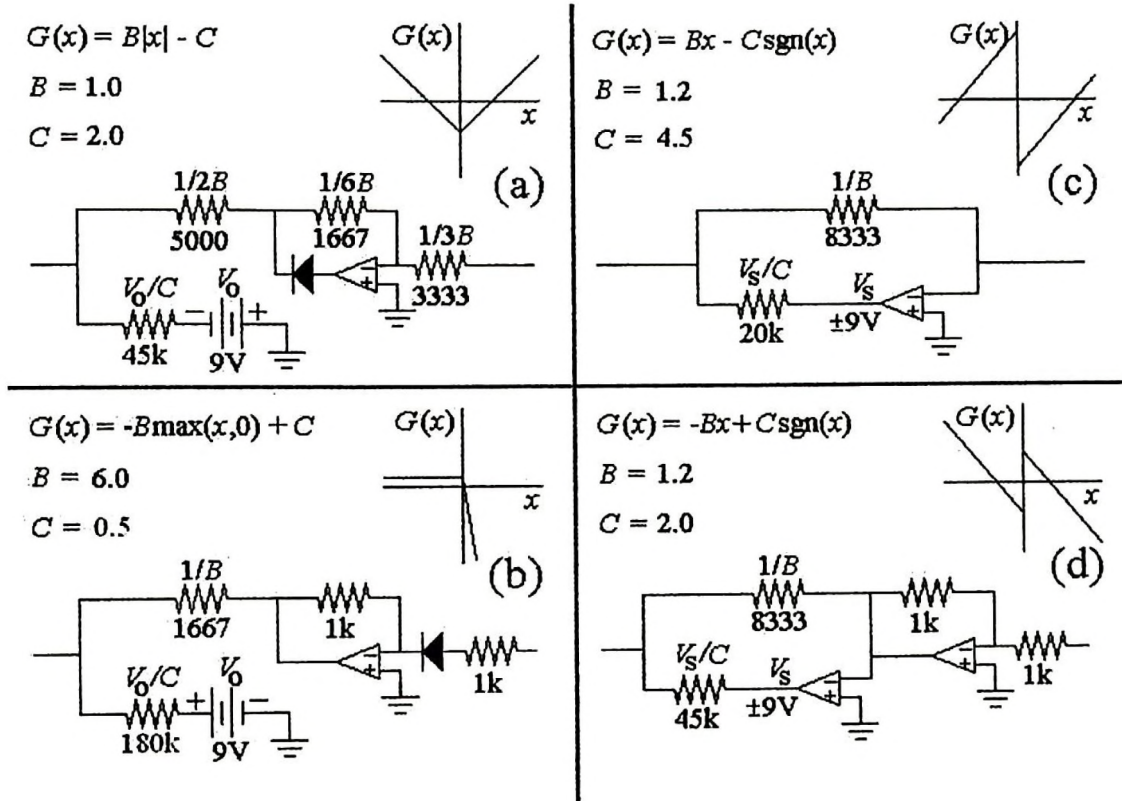


Figure 1.3: Some nonlinear forms for $G(x)$ that produce chaos and circuits to produce them. The parameter B is selected to be well within the chaotic regime. Constant C determines the size of the attractor and has been adjusted in each case so that x reaches a maximum value of slightly less than $5V$, a voltage well above noise level, but comfortably below a value that would typically saturate the op amps (Taken from reference [3]).

Chapter 2

Standard Nonlinear Dynamics Tools

2.1 Phase Space Trajectories

The evolution of a nonautonomous, time-continuous dynamical system is given by

$$\dot{\vec{x}} = \vec{F}(\vec{x}, t) \quad (2.1)$$

where the n -dimensional vector $\vec{x}$ is the state vector which describes the state of the system at any time t and the vector $\vec{F}$, often referred to as the vector field, is the evolution operator which determines how the system given by equation 2.1 will evolve in time. The space in which $\vec{x}$ evolves is known as the state space, otherwise called a phase space when half of the states are displacements and the other half are velocities. The dimension of a given system corresponds to the number of components of the state vector used to characterize the state of the system[13]. Given that we only need two state variables, $x(t)$ (displacement) and $v(t)$ (velocity), to characterize the state of the harmonic oscillator, we can say that it is a two dimensional system.

An orbit or trajectory is the projection of a solution $\vec{x}(t)$ of eqn. 2.1 onto the n -dimensional state space (Note: individual components $x_i(t)$ of $\vec{x}(t)$ are referred to as the system's time series). In other words, a solution of any dynamical system given by eqn. 2.1 can be thought of as a point that moves along a trajectory in much the same way that a planet moves through space. A point along this trajectory would specify the state of the system at that particular instant t . For systems with dimension $n \leq 2$, only periodic motion, limit cycles and fixed point behaviour can be observed, a consequence of the Poincaré-Bendixson theorem. These manifest in the state space as trajectories on or asymptotically approaching a closed curve for periodic motion and limit cycles, respectively. Stable fixed points are asymptotically approached by trajectories while trajectories diverge from unstable fixed points. If these fixed points are chosen as the system's initial condition, the system will stay fixed on these points as it evolves, hence the name.

More complex behaviour come in the form of bounded nonperiodic trajectories which visit regions of the state space while moving along geometric objects called ‘strange attractors’. Dynamical systems with strange attractors may or may not be chaotic, and it must be pointed out that visual inspection of the phase space plots is rarely sufficient for discerning quasiperiodic dynamics – that is, periodic solutions with at least 2 incommensurate frequencies – from chaotic dynamics. It may be easy to differentiate quasiperiodic dynamics from chaos if the former has few contributing frequencies or periodicities (indicated by a finite number of loops or orbits which can be visually resolved in the phase space plot) as opposed to the latter which has more loops or orbits than can be resolved by the eye. However, it sometimes happens that quasiperiodic dynamics can no longer be differentiated from chaos if the former possesses a very large number of contributing frequencies or loops in its phase space plot.

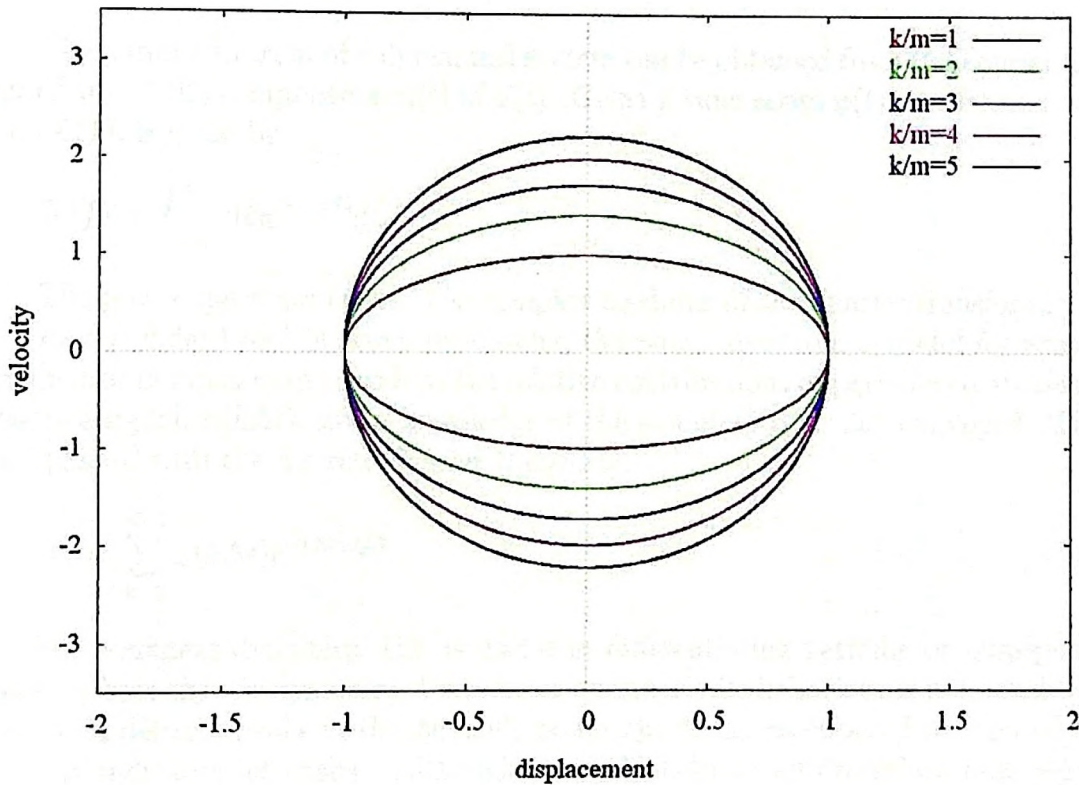


Figure 2.1: Phase space plots of the harmonic oscillator for different $\frac{k}{m}$ values.

As an example, consider the simple harmonic oscillator $\ddot{x} = -x$ with $\frac{k}{m} = 1$. Letting $x_1 = x$ (displacement) and $x_2 = \dot{x}$ (velocity), we obtain a form similar to equation 2.1,

given explicitly as

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_1 \end{bmatrix} \quad (2.2)$$

with the first order derivatives given by $\dot{\vec{x}} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix}$ and the evolution operator given by $\vec{F} = \begin{bmatrix} x_2 \\ x_1 \end{bmatrix}$. Typical elliptic trajectories corresponding to this system's periodic behaviour can be seen in figure 2.1 for different ratios of the parameters k (spring constant) and m (mass).

2.2 Power Spectrum

The power spectrum of a dynamical system can be obtained from the Fourier transform of any of the components $x_i(t)$ of $\vec{x}(t)$. Given a time series $x(t)$, the Fourier transform, $X(f)$, is given by

$$X(f) = \int_{-\infty}^{\infty} x(t) e^{-i2\pi f t} dt. \quad (2.3)$$

The power spectrum is then the complex modulus of the Fourier transform.

As a standard tool in signal processing, the power spectrum is useful for revealing periodicities in input data as well as the relative contributions of periodic components in a discrete signal, $x(k\Delta t)$, given knowledge of the sampling rate, Δt , employed. This is accomplished with the discrete Fourier transform,

$$X_n = \sum_{k=0}^{N-1} x(k\Delta t) e^{-i2\pi n k / N} \quad (2.4)$$

For nonlinear dynamics, this is useful in differentiating periodic or quasiperiodic dynamics from chaotic dynamics. Periodic or quasiperiodic behaviour is indicated by the presence of discrete peaks in the system's power spectrum, as opposed to a broad band spectrum indicative of chaos. Although it would help to autocorrelate quasiperiodic signals before finding their Fourier Transform, it is not strictly necessary if one only intends to verify the system's quasiperiodicity – versus chaotic behaviour in phase space plots as these two may look very much alike – without needing precise values of the contributing frequencies.

It must be noted, however, that the power spectrum alone cannot be used as a sufficient indicator of chaos in a dynamical system since noise also possesses broad band spectra.

2.3 Lyapunov Exponents

Lyapunov exponents provide a quantitative measure of a dynamical system's sensitivity to initial conditions. Given a bounded long-time evolution of a time-continuous dynamical system, negative exponents corresponding to exponential decrease in the separation of trajectories resulting from nearby initial states indicate limit cycle behaviour or quasiperiodicity. Chaos, marked by an exponential increase in the separation of trajectories with nearby initial conditions, is characterized by positive exponents. There are as many Lyapunov exponents in a dynamical system as there are state variables, but only one positive exponent is required for chaos.

Given an infinite amount of data, the i^{th} Lyapunov exponent corresponding to the i^{th} state variable in state space is given by

$$\lambda_i = \lim_{t \rightarrow \infty} \frac{1}{t} \log \frac{l_i(t)}{r_i(0)} \quad (2.5)$$

where $l_i(t)$ is the separation at time t of two trajectories starting with an initial separation $r_i(0)$.

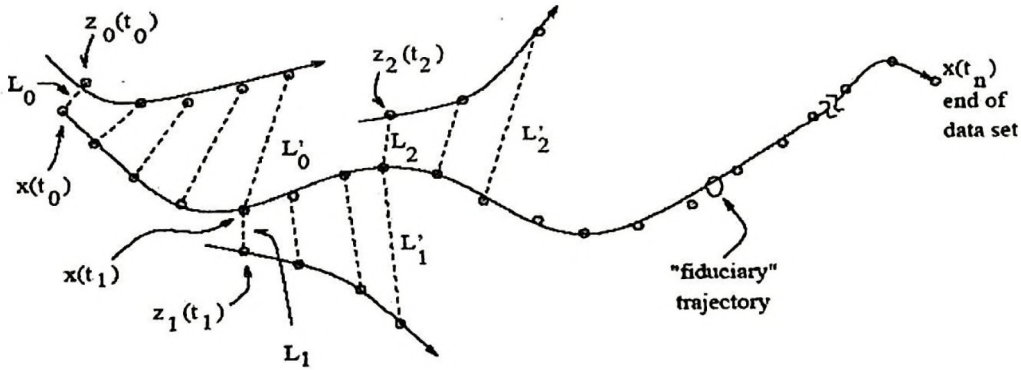


Figure 2.2: Graphic representation of the delay reconstructed orbit from a single time series (Taken from reference [9]).

In this thesis, it is computed using a publicly available algorithm in Fortran created by Wolf. The program codes, BASGEN and FET, make use of the time delay method of phase space reconstruction. BASGEN initially creates a database based on the input time series, after which FET calculates the Lyapunov exponents of the system from the reconstructed phase space. The $ndim$ -dimensional orbit and the time delay τ is used as input based on the estimated number of points for an orbit in the time series and the embedding dimension (recommended as $1/3$ of an orbit) one wants to use to reconstruct

the system, respectively. A multi-dimensional phase space is then reconstructed from a one-dimensional time series. From the database made by BASGEN, FET locates a pair of points from the reconstructed phase space that are very close to each other and follows the trajectory of these two points as they travel along the phase space. The separation is compared using the Euclidian distance of the two points. If the points grow very far apart, another segment is used by choosing another point within the phase space. Local rates of divergence/convergence of these points is averaged and divided by the total travel time, obtaining a long time average rate of divergence/convergence for nearby trajectories. A series of Lyapunov exponents is produced from a number of segments used within the reconstructed phase space. The last exponent in this series is the final Lyapunov exponent for the input time series.[6, 7]

In detail, Wolf's algorithm is implemented as follows[9, 7]. A graphic representation of the embedded trajectory is seen in figure 2.2 along with the relevant trajectory points mentioned in steps 2 to 6:

1. Starting with a time series $x = \{x_1, x_2, x_3, \dots, x_{n-2}, x_{n-1}, x_n\}$, we embed the data set via delay reconstruction method. For example, given an embedding dimension $d = 3$ and time delay $\tau = 5$, the delay-reconstructed orbit would consist of the sequence of points $(x_1, x_6, x_{11}), (x_2, x_7, x_{12}), (x_3, x_8, x_{13}), \dots$. In general, delay reconstruction produces the d -tuples $(x(t), x(t + \tau), x(t + 2\tau), \dots, x(t + (d - 1)\tau))$
2. We choose a point $x(t_0)$ somewhere in the middle of the trajectory.
3. We find that point's nearest neighbor and call it $z_0(t_0)$.
4. The Euclidian distance, $L_0 = \|z_0(t_0) - x(t_0)\|$, of the two points is then computed.
5. We then follow the "difference trajectory" – the dashed line – forward in time, compute $L_0(i) = \|z_0(t_i) - x(t_i)\|$ and increment i until $L_0(i) > \epsilon$, where ϵ is some tolerance, recommended by Wolf to be at around 15 percent of the data range. We call that value L'_0 and that time t_1 .
6. We then find $z_1(t_1)$, the "nearest neighbor" of $x(t_1)$, and loop to step 4. This procedure is repeated until the end of the fiduciary trajectory (i.e., $t = t_n$), while keeping track of the L_i and L'_i .

The formula used to compute the biggest Lyapunov exponent, λ_1 , is given by

$$\lambda_1 \approx \frac{1}{N\Delta t} \sum_i^{M-1} \log_2 \frac{L'_i}{L_i} \quad (2.6)$$

where M is the number of times one goes through the loop above and N is the number of time steps in the fiduciary trajectory. Also, $N\Delta t = t_n - t_0$.

Chapter 3

Sprott Circuits

In 1999, Linz and Sprott, in a search for the algebraically simplest chaotic flow with an absolute value nonlinearity, discovered the case

$$\frac{d^3x}{dt^3} + a\frac{d^2x}{dt^2} + b\frac{dx}{dt} - |x| + 1 = 0 \quad (3.1)$$

which exhibits chaos for $a = 0.6$ and $b = 1$. The system is also chaotic for reversed signs of the last two terms, exhibiting a mirror image of the original attractor about $x = 0$. The attractor itself resembles that in figure 3.1 with a folded-band structure.

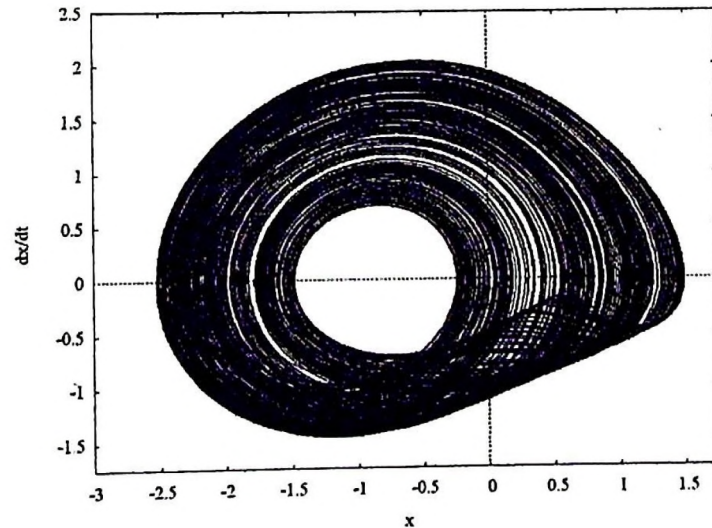


Figure 3.1: Phase space plot for one of the parameter pairs examined in the work that yield the ‘most chaotic’ solutions.

The constant 1 in equation 3.1 affects the size of the attractor. Although chaos still exists for arbitrarily small values of this constant, the attractor and its basin of attraction shrink proportionally. The authors both claim that it is the most elementary piecewise linear chaotic flow and point out that its piecewise linear nature allow for an analytic solution by solving two linear equations and matching the boundary conditions at $x = 0$. [2]

J. C. Sprott followed up on this chaotic flow with an investigation in 2000 of a new class of chaotic circuits of the general form

$$\frac{d^3x}{dt^3} + A\frac{d^2x}{dt^2} + \frac{dx}{dt} = G(x) \quad (3.2)$$

whose circuit implementation can be seen in figure 1.2, with some nonlinear forms for $G(x)$ shown in figure 1.3. Such equations often arise in the feedback control of an oscillator in which the experimentally accessible variable is a transformed and integrated version of the fundamental dynamical variable.

According to Sprott, equation 3.2 with a piecewise linear $G(x)$ suggests a class of chaotic circuits that are simple to construct, analyze and scale to most any desired frequency, being new in the sense that it doesn't involve analog multiplication, using only resistors, capacitors, diodes and operational amplifiers. Also, the governing equation is simpler than any previously modeled electronically.

Sprott's circuits are similar to Chua's circuit which use two capacitors, an inductor, and diodes with operational amplifiers or transistors to provide a piecewise linear approximation to a cubic nonlinearity. However, Chua's circuit has a much more complicated representation in terms of $\frac{d^3x}{dt^3}$ with more than four terms involving step functions, delta functions, and their products with derivatives of x . Also, Chua's circuit is much more difficult to construct, scale to arbitrary frequencies, and analyze because of the inductor with its frequency-dependent resistive losses. [3]

Chapter 4

Methodology

The nonlinear system being examined is given by

$$\frac{d^3x}{dt^3} = -A \frac{d^2x}{dt^2} - \frac{dx}{dt} + |x| - 1 \quad (4.1)$$

where A is an inverse resistance oscillating about a typical value $R_C = 1.67\Omega$ that gives rise to chaos [3, 4], modified in this work explicitly as

$$A = \frac{1}{R_C + R \sin(\omega t)} \quad (4.2)$$

where Sprott's original system is realized when $R = 0\Omega$, with Sprott's original circuit implementation given in figure 4.1. A possible implementation of our modification would be a replacement of the variable resistor in figure 4.1 with a constant resistor – corresponding to R_C – in series with a light dependent resistor or LDR – corresponding to $R \sin(\omega t)$ – modulated by a tunable flickering light source.

In our modified Sprott system, R at any one time is made to take on the discrete values

$$R = \{0.15\Omega, 0.10\Omega, 0.05\Omega, 0.025\Omega, 0.0125\Omega, 0.00625\Omega, 0.003125\Omega\} \quad (4.3)$$

and ω was made to vary with stepsize $\Delta\omega = 2^{-8}$ on the interval $[0, 20]$ for each value of R given in equation 4.3.

The solutions for equation 4.1 were obtained using a 4th order Runge-Kutta algorithm given by

$$x_{n+1} = x_n + \frac{1}{6} (k_1 + 2k_2 + 2k_3 + k_4) \quad (4.4)$$

with coefficients

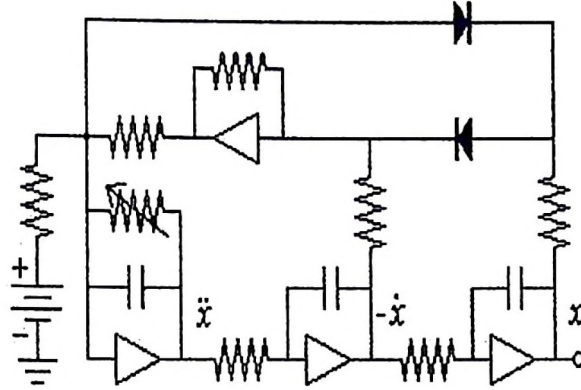


Figure 4.1: Chaotic circuit implementation of eqn. 4.1 with $R = 0\Omega$, using inverting op amps. The diodes are germanium, the battery is 1V, the capacitors are $0.1\mu F$, and the resistors are $1k\Omega$ except for the variable resistor – related to the parameter A in eqn. 4.1 via $R_{variable} = \frac{1}{A}$ – which should be adjustable from 1 to $2k\Omega$ (Taken from reference [4]).

$$\begin{aligned}
 k_1 &= \Delta t f(t_n, x_n) \\
 k_2 &= \Delta t f\left(t_n + \frac{\Delta t}{2}, x_n + \frac{k_1}{2}\right) \\
 k_3 &= \Delta t f\left(t_n + \frac{\Delta t}{2}, x_n + \frac{k_2}{2}\right) \\
 k_4 &= \Delta t f(t_n + \Delta t, x_n + k_3).
 \end{aligned}$$

The Runge-Kutta algorithm was used after eqn. 4.1 was converted to a set of first order ordinary differential equations given by

$$\begin{aligned}
 \frac{dx^{(1)}}{dt} &= x^{(2)} \\
 \frac{dx^{(2)}}{dt} &= x^{(3)} \\
 \frac{dx^{(3)}}{dt} &= -Ax^{(3)} - x^{(2)} + |x^{(1)}| - 1
 \end{aligned} \tag{4.5}$$

after introducing the variables

$$\begin{aligned}
x^{(1)} &= x \\
x^{(2)} &= \frac{dx}{dt} \\
x^{(3)} &= \frac{d^2x}{dt^2}.
\end{aligned}$$

The equations in 4.5 were iterated using eqn. 4.4 for 2^{15} time steps with a step size $\Delta t = 0.05$. The Lyapunov exponents of the solutions were then computed using Wolf's algorithm and plotted against the parameter ω , with a separate plot for each value of R from 4.3.

For visualization of the results, representative phase space plots were obtained and tabulated for the most chaotic solutions, least chaotic solutions and intermediate solutions, along with their corresponding power spectra and Lyapunov exponents.

Chapter 5

Results and Discussion

For clarity, we will refer to a set of parameters as the pair (R, ω) where R can be any of the values in 4.3 as explicitly stated. Unless otherwise stated, the parameter ω takes on all values on the interval $[0, 20]$.

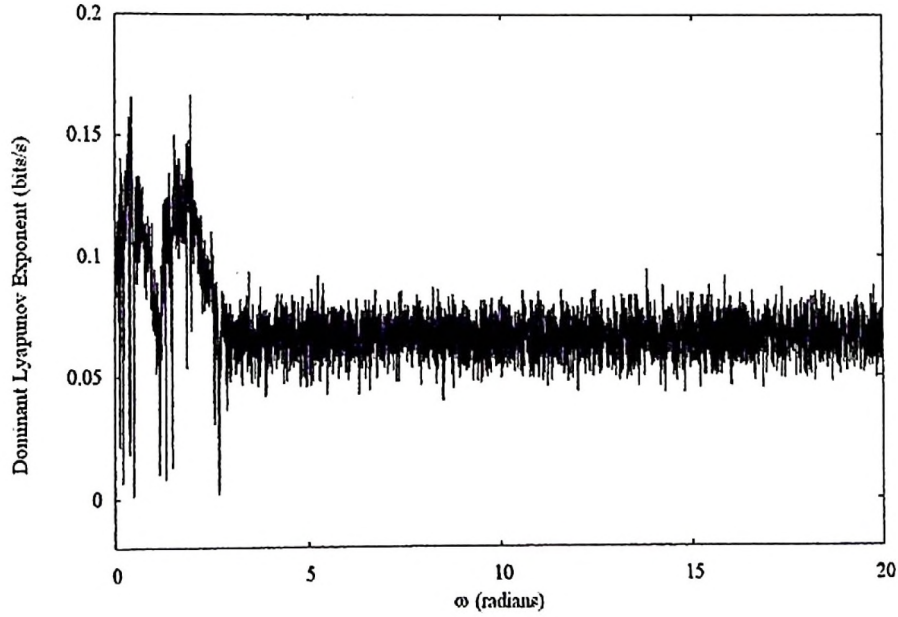


Figure 5.1: System dependence on ω for $(0.15\Omega, \omega)$ pair.

Initially, we apply the parameter pair $(0.15\Omega, \omega)$ to eqn. 4.1 to test the system's parameter dependence on ω . Our initial result can be seen in figure 5.1. It's notable that

at some parameter band $0 \leq \omega \leq 2.75$ radians, the dominant Lyapunov exponent (DLE) fluctuates erratically before eventually settling on some average value for higher ω values. We stopped computing the solutions for eqn. 4.1 at a cutoff value of $\omega = 20$ radians upon noting that further increase in this parameter resulted in little or no dramatic variation of the system's average DLE.

Reducing the parameter pair to $(0.05\Omega, \omega)$ and superimposing this result on the result obtained with figure 5.1, we observe in figure 5.2 a qualitatively similar fluctuation, but with smaller amplitude within the same parameter band $0 \leq \omega \leq 2.75$ radians and a lower average DLE for higher ω values. For clarity, we only plot the parameter dependence of the systems DLE up to $\omega = 6$ radians.

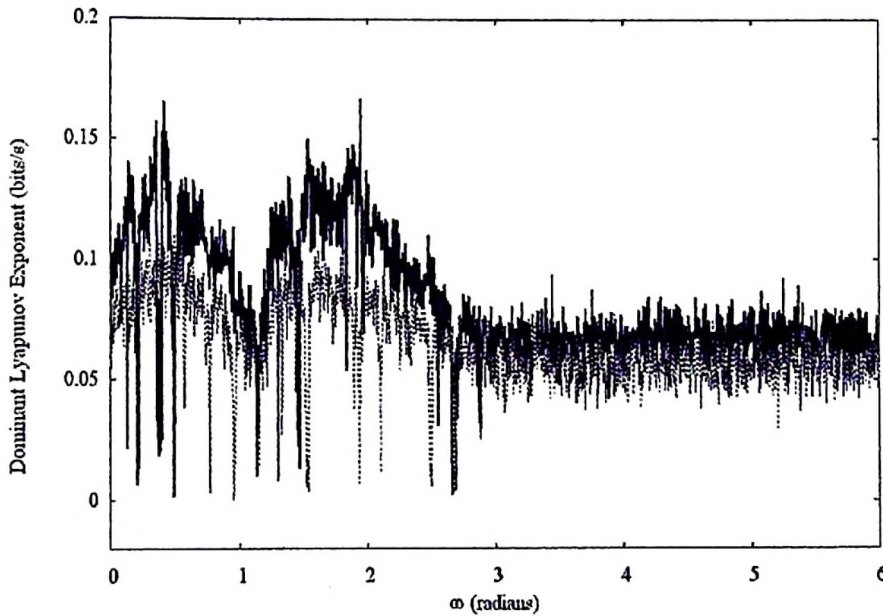


Figure 5.2: System dependence on ω for $(0.05\Omega, \omega)$ pair [dots], superimposed on $(0.15\Omega, \omega)$ pair [solid lines] for comparison.

To better observe how the system's DLE qualitatively depends on R , we smoothed our data by rendering it monotonic and afterwards approximating with a Bezier curve of degree N , corresponding to the number of data points. The result is seen in figures 5.4 and 5.5. We note in figure 5.4 that increasingly smaller values of R result in a smaller amplitude of fluctuation on the systems DLE, with the relative minimum and maxima preserved along the ω parameter axis regardless of R amplitude. Also worth noting in figure 5.5 is the observation that for $R < 0.05\Omega$, the average DLE values for higher ω no

longer decrease.

Representative phase space plots and power spectra for the parameter pair ($R = 0.15\Omega, \omega$) with arbitrary ω , together with their Lyapunov exponents, can be seen in tables 5.1 to ref 5.3 to better illustrate the system's level of chaoticity for those parameter pairs.

The system's Lyapunov exponent was also plotted as a function of the parameter R in the "averaging" regime (i.e., large ω) and can be seen in figure 5.3. A general trend of increasing chaoticity or Lyapunov exponent value was found for increasing R values.

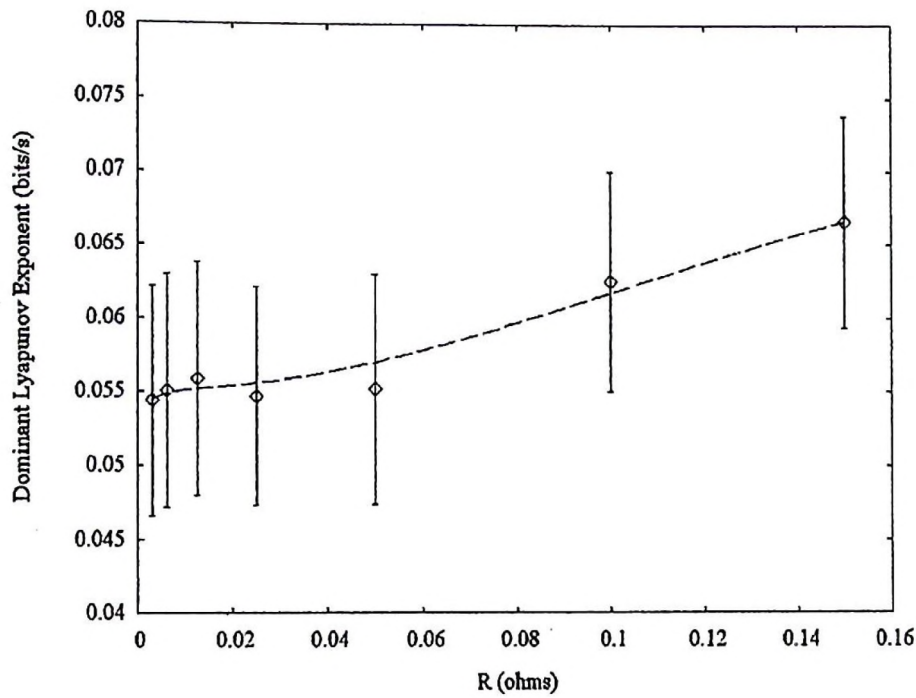


Figure 5.3: System dependence on R for large ω .

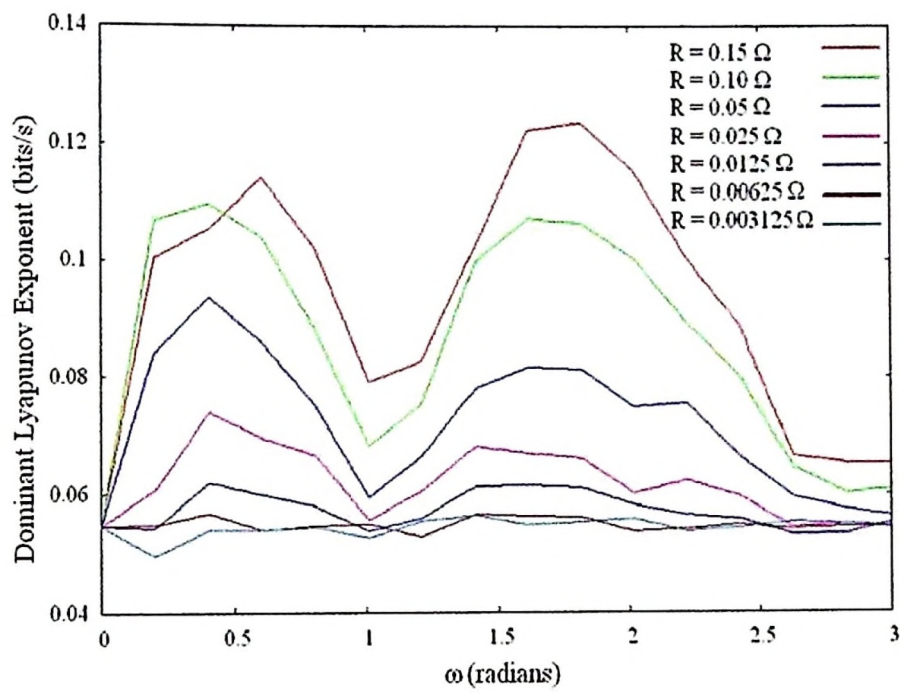


Figure 5.4: Enlarged view of the system's varying chaoticity as a function of ω for small ω values.

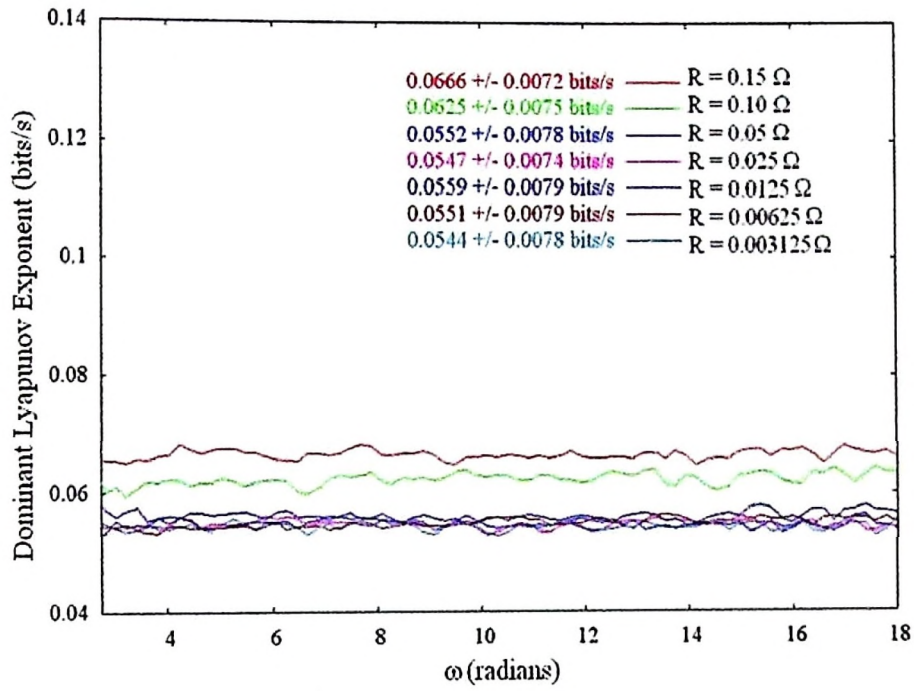


Figure 5.5: "Averaging behaviour" of the system for large ω values.

Table 5.1: Phase Space Plots of Increasing Chaoticity for $R = 0.15\Omega$ with Corresponding Power Spectrum and Lyapunov exponents

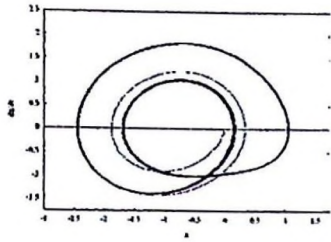
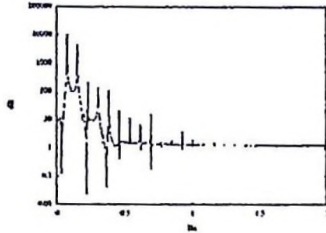
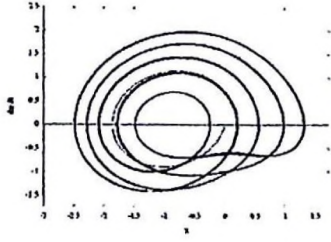
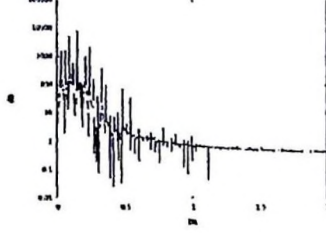
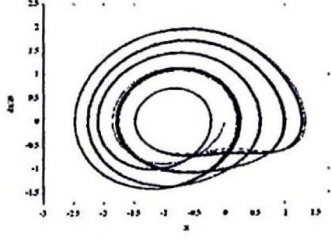
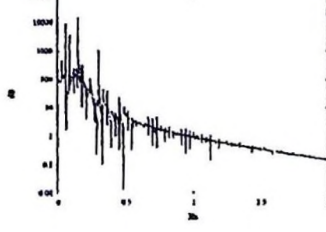
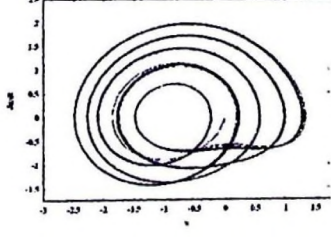
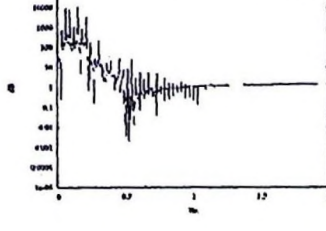
ω (radians)	Phase Space Trajectory	Power Spectrum	LE (bits/s)
0.48828125			0.0016
2.6640625			0.0027
2.68359375			0.0039
2.6796875			0.0045

Table 5.2: Phase Space con't...

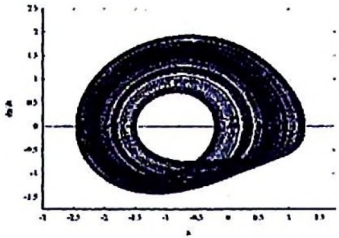
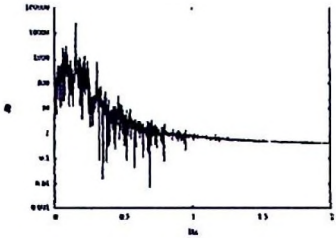
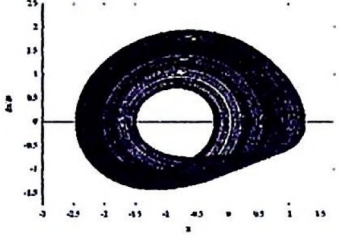
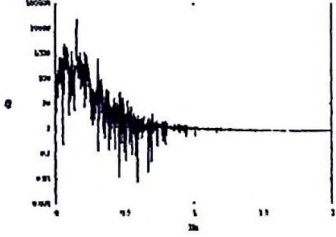
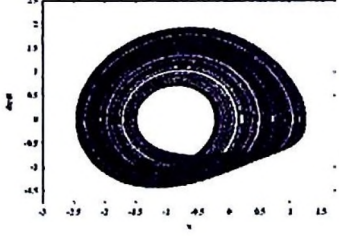
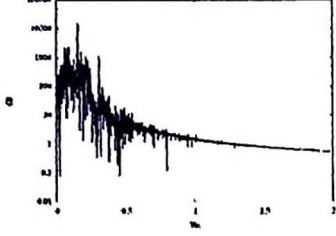
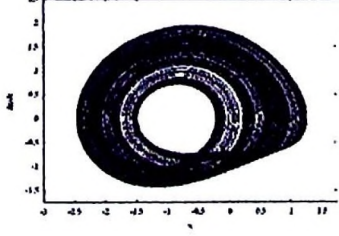
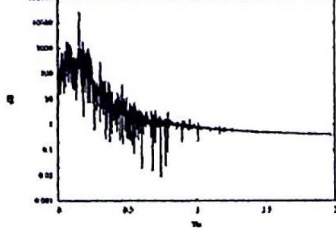
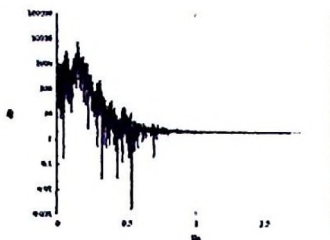
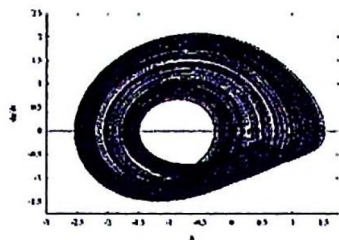
ω (radians)	Phase Space Trajectory	Power Spectrum	LE (bits/s)
19.46875			0.071
15.53125			0.0711
7.13671875			0.0713
15.1640625			0.0714

Table 5.3: Phase Space con't...

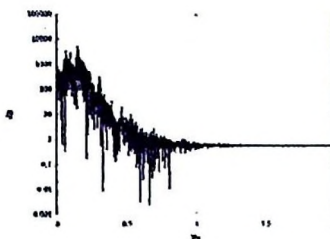
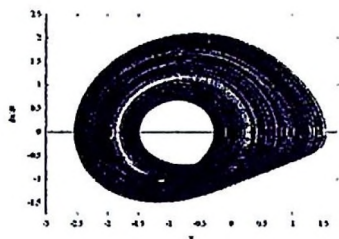
ω (radians)	Phase Space Trajectory	Power Spectrum	LE (bits/s)
--------------------	------------------------	----------------	-------------

0.4375



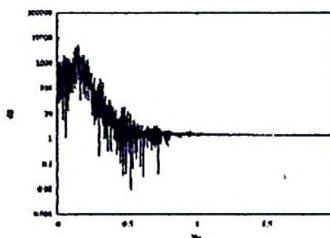
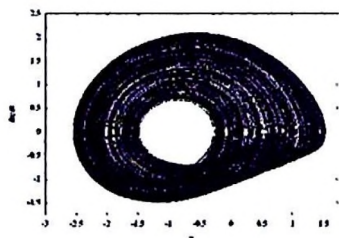
0.0752

0.31640625



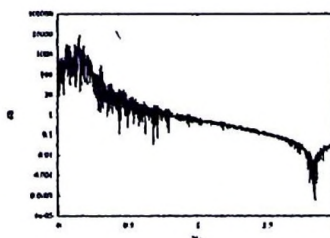
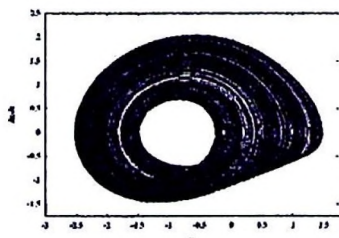
0.1393

0.41015625



0.1597

1.94921875



0.1667

Chapter 6

Conclusions and Recommendations

In this thesis, unexpected qualitative effects on the behaviour of a particular known chaotic Sprott circuit subject to numerical investigation were observed to manifest as parameter values of an additive resistance were varied.

Numerical solutions of the system's ODE obtained via a standard 4th order Runge-Kutta algorithm and subjected to Wolf's algorithm to obtain its corresponding Lyapunov exponents yielded a fluctuating DLE for the system when the parameter ω is within the range $0 \leq \omega \leq 2.75$ radians. Beyond this parameter range, the system was observed to assume an "averaging behaviour" in which the system's DLE assumed an approximately constant value upon further increase on ω .

Although fluctuations on the system's Lyapunov exponents within the parameter range $0 \leq \omega \leq 2.75$ radians were observed to persist even for decreasing values of R , this was observed to die out eventually as the system approaches the original Sprott configuration with $R = 0\Omega$.

Also worth noting in the system's behaviour was the persistence of an average DLE value for $R < 0.05\Omega$ at higher ω values (i.e., $\omega > 2.75$ radians).

A realization of the modification introduced in this thesis may be implemented by using a variable resistor that can be tuned with the numerically applied frequencies using some external parameter. An LDR may be feasible if exposed to a light source connected to a function generator. However, further work may be needed in terms of adjusting this parameter in order to implement practical frequency values the way Sprott has done in scaling the voltage frequencies up into the audio range.

Autocorrelating signals before finding their Fourier Transform may not necessarily provide a significant improvement in the efficacy of the power transform for differentiating quasiperiodic signals from chaotic signals, since one only needs to see that the latter has a broadband spectra while the former has a discrete spectra. However, if one needs precise values of the contributing frequencies in the case of quasiperiodic signals, autocorrelation is, indeed, a necessary preliminary step.

Analytical verification of the observed “averaging behaviour” may also be realized by using a multiple timescales treatment, where one may assume a $\tau_1 \gg t$ and $\tau_2 \ll t$ where $\tau_1 = \omega_1 t$ and $\tau_2 = \omega_2 t$ for large and small ω values, respectively.

In summary, we have obtained numerical results on the dynamics of a modified Sprott circuit, proposed an actual implementation of the modification we introduced, and proposed an analytical verification of the “averaging behaviour” observed at large values of the parameter ω .

Appendix A

BASGEN (dataBASE Generator)

```
program BASGEN
C   INPUT: 4column data in file "data" w/ 1st column as input time series
parameter(maxdat=32768,maxbox=6000,maxdim=8)
integer*4 nxtbox(maxbox,maxdim),where(maxbox,maxdim)
integer*4 datptr(maxbox),nxt-dat(maxdat),target(maxdim)
integer*4 tau,datcnt,boxcnt,runner,chaser,used
dimension data(maxdat),tdata(maxdat,4)
real mindata,maxdata
logical tru
C   storage requirements: 2*maxdat + maxbox*(2*maxdim+1)
write(*,111)
111 format(1x,'ASCII data file=1 : ')
   iform = 1
   write(*,*) iform
   write(*,222) maxdat
222 format(1x,'number of data points (<=',i7,') : ')
   datcnt = maxdat
   write(*,*) datcnt
   open(unit=1,file='data',status='old')
   do i=1,datcnt
     read(1,*) (tdata(i,j),j=1,4)
   end do
   close(1)
   do i=1,datcnt
     data(i) = tdata(i,1)
   end do
   if(iform.eq.1) then
```



```

        inquire(file='t.',exist=tru)
        if(tru) then
            open(unit=2,file='t.')
            close(unit=2,status='delete')
        endif
        open(unit=2,file='t.',form='unformatted')
        do 10 i=0,(datcnt/8)-1
            write(2) (data(8*i+j),j=1,8)
10         continue
            close(unit=2)
        else
            open(unit=1,file='t.',status='old',form='unformatted')
            do 20 i=0,(datcnt/8)-1
                read(1) (data(8*i+j),j=1,8)
20             continue
                close(unit=1)
            endif
            datmin=data(1)
            datmax=data(1)
            do 30 i=1,datcnt
                datmin=amin1(datmin,data(i))
                datmax=amax1(datmax,data(i))
30             continue
C         ENLARGE GRID SO NO POINTS ON OUTER BOUNDARY
            mindata=datmin
            maxdata=datmax
            datmin=datmin-.01*(datmax-datmin)
            datmax=datmax+.01*(datmax-datmin)
40         write(*,333)
333        format(1x,'time delay (samples)  : ')
            tau = 15
            write(*,*) tau
            write(*,444) maxdim
444        format(1x,'embedding dimension (<=',i4,')  : ')
            ndim = 4
            write(*,*) ndim
            write(*,555) maxbox
555        format(1x,'grid resolution (maxbox =',i7,')  : ')
            ires = 10
            write(*,*) ires

```

```

        write(*,*) 'Max data value: ',maxdata
        write(*,*) 'Min data value: ',mindata
        boxlen=(datmax-datmin)/ires
C      INITIALIZE THE GRID
        boxcnt=1
        do 60 i=1,maxbox
            datptr(i)=0
            do 50 j=1,ndim
                nxtbox(i,j)=0
                where(i,j)=0
50          continue
60        continue
        do 70 i=1,maxdat
            ntxdat(i)=0
70        continue
C      MAIN LOOP
        do 140 i=1,datcnt-(ndim-1)*tau
C      DETERMINE TARGET COORDINATES
            do 80 j=1,ndim
                target(j)=(data(i+(j-1)*tau)-datmin)/boxlen
80          continue
C      GO TO TARGET BOX, DIMENSION BY DIMENSION
C      IF BOX DOESN'T EXIST, CREATE IT
            runner=1
            chaser=0
            do 130 j=1,ndim
90              if(where(runner,j)-target(j)) 100,130,110
100             chaser=runner
                runner=nxtbox(runner,j)
                if(runner.ne.0) goto 90
110            boxcnt=boxcnt+1
                if(boxcnt.eq.maxbox) then
C      TOO MANY BOXES - QUIT
                    write(*,*) 'grid overflow at i = ',i
                    goto 40
                endif
                do 120 k=1,ndim
                    where(boxcnt,k)=where(chaser,k)
120              continue
                where(boxcnt,j)=target(j)

```

```

        nxtbox(chaser,j)=boxcnt
        nxtbox(boxcnt,j)=runner
        runner=boxcnt
130    continue
C    RUNNER IS TARGET BOX - ADD DATA POINT TO THIS BOX
        nxtdat(i)=datptr(runner)
        datptr(runner)=i
140    continue
        used=0
        do 150 i=1,boxcnt
            if(datptr(i).ne.0) used=used+1
150    continue
        write(*,*) '# boxes allocated : ',maxbox
        write(*,*) '# boxes created   : ',boxcnt
        write(*,*) '# boxes non-empty : ',used
        write(*,*)
        write(*,666)
666    format(1x,'ok=1 : ')
        iok = 1
        write(*,*) iok
        if(iok.ne.1) goto 40
C    WRITE OUT DATA BASE
C    (EMPTY BOXES WRITTEN OUT - WASTES SOME STORAGE SPACE)
        inquire(file='database.',exist=tru)
        if(tru) then
            open(unit=1,file='database.')
            close(unit=1,status='delete')
        endif
        open(unit=1,file='database.',form='unformatted')
        write(*,*) 'writing the database...'
        write(1) ndim,ires,tau,boxcnt,datcnt,datmax,datmin,maxdata,mindata
1    ,boxlen
        do 160 i=1,boxcnt
            write(1) datptr(i)
            write(1) (nxtbox(i,j),j=1,ndim)
            write(1) ( where(i,j),j=1,ndim)
160    continue
        do 170 i=1,datcnt
            write(1) nxtdat(i)
170    continue

```



```
close(unit=1)
end !end program BASGEN
```

Appendix B

FET (Fixed Evolution Time)

```
program FET
parameter(maxdat=32768,maxdim=8,maxbox=6000)
common ndim,ires,tau,boxcnt,datcnt,datmax,datmin,boxlen,nxtbox,
1  where,datptr,nxtat,data,delay,oldpnt,newpnt,bstpnt,
2  runner,datuse,dismin,dismax,bstdis,thmax,thbest,evolve
dimension data(maxdat)
integer*4 nxtbox(maxbox,maxdim),where(maxbox,maxdim)
integer*4 datptr(maxbox),nxtat(maxdat),delay(maxdim)
integer*4 tau,boxcnt,datcnt,datuse,evolve
integer*4 oldpnt,bstpnt,runner
real maxdata,mindata
logical tru
C READ IN THE DATA BASE
open(unit=1,file='database.',status='old',form='unformatted')
write(*,*)
write(*,*) 'reading the database...'
read(1) ndim,ires,tau,boxcnt,datcnt,datmax,datmin,maxdata,mindata
1 ,boxlen
do 10 i=1,boxcnt
  read(1) datptr(i)
  read(1) (nxtbox(i,j),j=1,ndim)
  read(1) ( where(i,j),j=1,ndim)
10 continue
do 20 i=1,datcnt
  read(1) nxtat(i)
20 continue
close(unit=1)
```

```

        write(*,*) 'ndim      : ',ndim
        write(*,*) 'ires      : ',ires
        write(*,*) 'tau       : ',tau
        write(*,*) 'boxcnt    : ',boxcnt
        write(*,*) 'datcnt    : ',datcnt
        write(*,*) 'datmax    : ',datmax
        write(*,*) 'datmin    : ',datmin
        write(*,*) 'boxlen    : ',boxlen
        write(*,*) 'Max data value: ',maxdata
        write(*,*) 'Min data value: ',mindata
C READ IN THE TIME SERIES
        open(unit=1,file='t.',status='old',form='unformatted')
        write(*,*) 'reading the time series...'
        do 30 i=1,datcnt/8
            read(1) (data(8*(i-1)+j),j=1,8)
30      continue
        close(unit=1)
C ECHO PARAMETERS
        write(*,111)
111      format(1x,'time-step (seconds or iterations) : ')
        dt = 0.0500000000000000000000000000000000
        write(*,*) dt
        write(*,222)
222      format(1x,'evolution time (number of samples) : ')
        evolve = 9
        write(*,*) evolve
        write(*,333)
333      format(1x,'minimum separation at replacement : ')
        dismin = 0.02 * (maxdata-mindata)
        write(*,*) dismin
        write(*,444)
444      format(1x,'maximum separation for replacement : ')
        dismax = 0.15 * (maxdata-mindata)
        write(*,*) dismax
        write(*,555)
555      format(1x,'maximum orientation error          : ')
        thmax = 30
        write(*,*) thmax
        inquire(file='fet.out',exist=tru)
        if(tru) then

```



```

        open(unit=1,file='fet.out')
        close(unit=1,status='delete')
    endif
    open(unit=1,file='fet.out',status='new')
    do 40 i=1,ndim
        delay(i)=(i-1)*tau
40    continue
C some data points at the end are lost...
    datuse=datcnt-(ndim-1)*tau-evolve
    its=0
    sum=0.0
    savmax=dismax
    oldpnt=1
    newpnt=1

C SEARCH FOR NEAREST NEIGHBOR - IGNORE ORIENTATION
50    call search(0)
    if(bstpnt.eq.0) then
        dismax=dismax*2.0
        goto 50
    endif
    dismax=savmax
    newpnt=bstpnt
    disold=bstdis
    iang=-1

C EVOLVE POINTS, FIND FINAL SEPARATION, UPDATE EXPONENT
60    oldpnt=oldpnt+evolve
    newpnt=newpnt+evolve
C CALCULATION FINISHED
    if(oldpnt.ge.datuse) goto 110
C FELL OFF TIME SERIES - IN DESPERATION - FIND NEAREST NEIGHBOR
    if(newpnt.ge.datuse) then
        oldpnt=oldpnt-evolve
        goto 50
    endif
    disnew=0.0
    do 80 i=1,ndim
        p1=data(oldpnt+delay(i))
        p2=data(newpnt+delay(i))
        disnew=disnew+(p2-p1)**2.

```

```

80    continue
      disnew=sqrt(disnew)
      if (disnew.lt.0.1*dismin) disnew=0.1*dismin
      its=its+1
      sum=sum+alog(disnew/disold)
      zlyap=sum/(its*evolve*dt*alog(2.0))
      if(iang.eq.-1) then
        write(1,777) evolve*its,disold,disnew,zlyap
      else
        write(1,888) evolve*its,disold,disnew,zlyap,iang
      endif
777  format(1x,i8,6x,f12.4,3x,f12.4,8x,f10.4)
888  format(1x,i8,6x,f12.4,3x,f12.4,8x,f10.4,8x,i5)
C    LOOK FOR REPLACEMENT ONLY IF FINAL DISTANCE > DISMAX
      if(disnew.le.dismax) goto 90
      call search(1)
      if(bstpnt.ne.0) goto 100
C    NO ACCEPTABLE REPLACEMENT - TAKE NEAREST NEIGHBOR
      goto 50
C    NO NEED FOR REPLACEMENT - KEEP EVOLVING
90    disold=disnew
      iang=-1
      goto 60
C REPLACEMENT SUCCESSFUL - START NEW EVOLUTION
100   newpnt=bstpnt
      disold=bstdis
      iang=int(thbest)
      goto 60
110   continue
      end !end program FET

      subroutine search(iflag)
      parameter(maxdat=32768,maxdim=8,maxbox=6000)
      common ndim,ires,tau,boxcnt,datcnt,datmax,datmin,boxlen,nxtbox,
1  where,datptr,nxtdat,data,delay,oldpnt,newpnt,bstpnt,
2  runner,datuse,dismin,dismax,bstdis,thmax,thbest,evolve
      dimension data(maxdat),oldcrd(maxdim),zewcrd(maxdim),
1  bstcrd(maxdim)
      integer*4 nxtbox(maxbox,maxdim),where(maxbox,maxdim)
      integer*4 datptr(maxbox),nxtdat(maxdat),delay(maxdim)

```

```

integer*4 igcrds(maxdim),target(maxdim)
integer*4 tau,boxcnt,datcnt,datuse,evolve
integer*4 oldpnt,bstpnt,runner
C IF BSTPNT <> 0 ON RETURN - A REPLACEMENT HAS BEEN FOUND.
C ...BSTPNT POINTS TO IT
C ...THBEST IS THE CHANGE IN ORIENTATION
C ...BSTDIS IS THE NEW SEPARATION FROM THE FIDUCIAL POINT
C ...BSTCRD HAS ITS COORDINATES (NOT PASSED BACK TO "MAIN")
C SET UP GRID COORDINATES
do 10 i=1,ndim
  oldcrd(i)=data(oldpnt+delay(i))
  zewcrd(i)=data(newpnt+delay(i))
  igcrds(i)=(oldcrd(i)-datmin)/boxlen
10 continue
C FIND DISTANCE FROM OLDPNT TO NEWPNT
C (NOT NECESSARILY 'DISNEW' OF MAIN PROGRAM)
oldist=0.0
do 20 i=1,ndim
  abc=oldcrd(i)-zewcrd(i)
  oldist=oldist+abc**2.
20 continue
oldist=sqrt(oldist)
C SKIP BOXES CLOSER THAN DISMIN, INITIALIZE 'BEST' VARIABLES
irange=int(dismin)/boxlen
if(irange.eq.0) irange=1
thbest=thmax
bstdis=dismax
bstpnt=0
30 do 140 icnt=0,((2*irange+1)**ndim)-1
  icounter=icnt
  do 40 i=1,ndim
    ipower=(2*irange+1)**(ndim-i)
    ioff=icounter/ipower
    icounter=icounter-ioff*ipower
    target(i)=igcrds(i)-irange+ioff
    if(target(i).lt.0) goto 140
    if(target(i).gt.i-res-1) goto 140
40 continue
C CHECK ALL BOXES ON OUTER PERIMETER OF SEARCH AREA
C (EXCEPT FIRST TIME THROUGH)

```



```

        if(irange.eq.1) goto 60
        iskip=1
        do 50 i=1,ndim
            if(iabs(target(i)-igcrds(i)).eq.irange) iskip=0
50      continue
            if(iskip.eq.1) goto 140
C      SEARCH FOR TARGET BOX
60      runner=1
        do 80 i=1,ndim
70      if(where(runner,i).eq.target(i)) goto 80
            runner=nxtbox(runner,i)
            if(runner.ne.0) goto 70
            goto 140
80      continue
            if(runner.eq.0) goto 140
            runner=datptr(runner)
            if(runner.eq.0) goto 140
C      FOR POINTS IN THIS BOX CHECK...
C      DISTANCE, ANGLE, NON-ADJACENCY
90      if(iabs(runner-oldpnt).lt.evolve) goto 130
            if(iabs(runner-datuse).lt.(2*evolve)) goto 130
            do 100 i=1,ndim
                bstcrd(i)=data(runner+delay(i))
100     continue
            tdist=0.0
            dot=0.0
            do 110 i=1,ndim
                abc1=oldcrd(i)-bstcrd(i)
                abc2=oldcrd(i)-zewcrd(i)
                tdist=tdist+abc1*abc1
                dot=dot+abc1*abc2
110     continue
            tdist=sqrt(tdist)
            if(tdist.lt.dismin) goto 130
            if(tdist.ge.bstdis) goto 130
            if(tdist.eq.0) goto 130
            if(iflag.eq.0) goto 120
            ctheta=amin1(abs(dot/(tdist*oldist)),1.0)
            theta=57.3*acos(ctheta)
            if(theta.ge.thbest) goto 130

```

```
      thbest=theta
120    bstdis=tdist
      bstpnt=runner
C     GO TO NEXT POINT IN BOX
130    runner=nxt-dat(runner)
      if(runner.ne.0) goto 90
140    continue
C RETURN IF POINT FOUND, ELSE CHECK NEXT LAYER
C NOTE DEPENDENCE OF # OF BOXES SEARCHED ON DISMAX
C DISMAX < 1.5*BOXLEN -> 3**NDIM BOXES (1 LAYER)
C DISMAX < 2.5*BOXLEN -> 5**NDIM BOXES (2 LAYERS)   ETC...
      irange=irange+1
      if(irange.le.int(0.5+(dismax/boxlen))) goto 30
      return
      end !end subroutine search
```

Appendix C

FFT (Fast Fourier Transform)

Here is the FFT code taken from reference [14]:

```
subroutine FFT(xreal,ximag,n,nu)
real*8 xreal(n),ximag(n)
real*8 arg,c,s,treal,timag
integer n,nu
integer k1n2,L,i,n2,nu1,k,k1,p
integer ibitr
external ibitr
n2=n/2
nu1=nu-1
k=0
do l=1,nu
102  do i=1,n2
      p=ibitr(k/2**nu1,nu)
      arg=6.283185d0*p/dbble(n)
      c=cos(arg)
      s=sin(arg)
      k1=k+1
      k1n2=k1+n2
      treal=xreal(k1n2)*c+ximag(k1n2)*s
      timag=ximag(k1n2)*c-xreal(k1n2)*s
      xreal(k1n2)=xreal(k1)-treal
      ximag(k1n2)=ximag(k1)-timag
      xreal(k1)=xreal(k1)+treal
      ximag(k1)=ximag(k1)+timag
      k=k+1
```



```

    end do
    k=k+n2
    if(k.lt.n) goto 102
    k=0
    nu1=nu1-1
    n2=n2/2
end do
do k=1,n
    i=ibitr(k-1,nu)+1
    if(i.le.k) goto 103
    treal=xreal(k)
    timag=ximag(k)
    xreal(k)=xreal(i)
    ximag(k)=ximag(i)
    xreal(i)=treall
    ximag(i)=timag
103 continue
end do
end !end subroutine FFT

integer function ibitr(j,nu)
integer j1,j,nu,j2
j1=j
ibitr=0
do i=1,nu
    j2=j1/2
    ibitr=ibitr*2+(j1-2*j2)
    j1=j2
end do
end !end function ibitr

```

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